MILITARY SPECIFICATION

CASE SECTION, BALLISTIC, CENTER, AFT

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of ballistic center section entitled as Case, Section, Ballistic, Center, Aft which is used in the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-48512 - Shipping and Storage Container, Atomic Warhead Section: M480.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

ORDNANCE CORPS

8822030	- Box, Packing, Ammunition for Case Section, Assembly Ballistic, Center, Aft.
8825072	- Case Section, Ballistic, Center Aft.
8825095	- Skin, Middle Section.
8825153	- Adaption Kit, Atomic Warhead to Rocket; M86.
8839515	- Shipping and Storage Container, Atomic Warhead Section: M480.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

8826411 - Warhead Section, Atomic: M27.
8831129 - Mark 31 Warhead, Compatibility Information.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The assembly shall comply with all requirements specified on the drawing (dwg) 8825072, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Mating assembly.

3.5.1 Adaption kit assembly. The case section, ballistic, center, aft shall be capable of mating mechanically and electrically when attached to mating assemblies shown in dwg 8825153 or their equivalents when tested as specified in 4.4.1.

3.5.2 Warhead compatibility. The case section, ballistic, center, aft shall be capable of mating mechanically with the applicable mating assemblies shown on dwg 8831129 or their equivalents. The test shall be conducted as specified in 4.4.2.

3.6 Resistance. The resistance of each circuit shall not exceed the maximum value listed in Table I when tested in accordance with 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Table I

Resistance Between Pin and Pin	Resistance (ohms)
W2K-P4-A to W2K-P2-E	0.05 max
W2K-P4-B to W2K-P2-G	0.05 max
W2K-P4-C to W2K-P2-K	0.05 max
W2K-P4-C to W2K-P2-L	0.05 max
W2K-P4-D to W2K-P2-P	0.05 max
W2K-P4-E to W2K-P2-S	0.05 max
W2K-P4-F to W2K-P2-V	0.05 max
W2K-P4-F to W2K-P2-W	0.05 max
W2K-P4-G to W2K-P1-A	0.05 max
W2K-P4-G to W2K-P1-T	0.05 max
W2K-P4-T to W2K-P1-F	0.05 max
W2K-P4-T to W2k-P1-V	0.05 max
W2K-P4-W to W2k-P1-Y	0.05 max
W2K-P4-X to W2k-P1-X	0.05 max
W2K-P4-Y to W2k-P2-D	0.05 max
W2K-P4-Z to W2k-P1-P	0.05 max
W2K-P1-B to W2K-P2-k	0.05 max
W2K-P1-C to W2K-P2-$\overline{J}$	0.05 max
W2K-P1-D to W2K-P2-$\overline{X}$	0.05 max
W2K-P1-E to W2K-P2-H	0.05 max
W2K-P1-G to W2K-P2-p	0.05 max
W2K-P1-H to W2K-P2-$\overline{d}$	0.05 max
W2K-P1-J to W2K-P2-$\overline{c}$	0.05 max
W2K-P1-K to W2K-P2-$\overline{T}$	0.05 max
W2K-P1-L to W2K-P2-A	0.05 max
W2K-P1-M to W2K-P2-B	0.05 max
W2K-P1-N to W2K-P2-C	0.05 max
W2K-P1-T to W2K-P4-G	0.05 max
W2K-P1-U to W2K-P2-M	0.05 max
W2K-P1-V to W2K-P4-T	0.05 max
W2K-P1-W to W2K-P4-i	0.05 max
W2K-P3-A to W2K-P5-$\overline{A}$	0.05 max
W2K-P3-B to W2K-P5-B	0.05 max
W2K-P3-C to W2K-P5-D	0.05 max
W2K-P3-D to W2K-P5-F	0.05 max
W2K-P3-E to W2K-P5-C	0.05 max
W2K-P3-F to W2K-P6-A	0.05 max
W2K-P3-G to W2K-P6-B	0.05 max
W2K-P3-H to W2K-P6-C	0.05 max

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Table I (Con't)

Resistance Between Pin and Pin	Resistance (ohms)
W3K-P2-A to W3K-P1-A	0.05 max
W3K-P2-B to W3K-P1-E	0.05 max
W3K-P2-C to W3K-P1-B	0.05 max
W3K-P2-E to W3K-P1-C	0.05 max
W3K-P2-F to W3K-P1-D	0.05 max
W3K-P2-G to W3K-P1-L	0.05 max
W3K-P1-M to W3K-P3-A	0.05 max
W3K-P1-K to W3K-P3-C	0.05 max
W2K-P3-J to W2K-P5-G	0.05 max
W3K-P1-F to W3K-P1-G	0.10 max
W3K-P1-F to W2K-P3-K	0.10 max
W2K-P3-N to W2K-P3-I	0.10 max
W2K-P3-N to W3K-P1-J	0.10 max
W2K-P3-L to W2K-P3-M	0.10 max
W2K-P3-L to W3K-P1-H	0.10 max
Ring fuze No. 2, Section No. 1	Ground (external skin)
Ring fuze No. 2, Section No. 2	Ground (external skin)
Ring fuze No. 2, Section No. 3	Ground (external skin)
Ring fuze No. 3, Section No. 1	Ground (external skin)
Ring fuze No. 3, Section No. 2	Ground (external skin)
Ring fuze No. 4, Section No. 1	Ground (external skin)
Ring fuze No. 4, Section No. 2	Ground (external skin)
Ring fuze No. 4, Section No. 3	Ground (external skin)
Ring fuze No. 4, Section No. 4	Ground (external skin)

3.7 Insulation resistance. The insulation resistance shall be 20 megohms minimum (min) between the points specified in Table II when subjected to 500 ± 50 volts direct current (vdc). Prior to testing, the following pins W2K-P2-k, p, -E, -P shall be subjected to 1500 ± 100 volts alternating current (vac) at 60 Hz between the terminals and ground (external skin). The assembly shall be tested as specified in 4.4.4.

Table II

Resistance Between Each Connector Pin and All Others Listed:	Resistance (megohms)
W2K-P4-A	20 min
W2K-P4-B	20 min
W2K-P4-C	20 min
W2K-P4-D	20 min
W2K-P4-E	20 min
W2K-P4-F	20 min
W2K-P4-G	20 min

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Table II (Con't)

Resistance Between Each Connector Pin and All Others Listed:	Resistance (megohms)
W2K-P4-J	20 min
W2K-P4-K	20 min
W2K-P4-Q	20 min
W2K-P4-T	20 min
W2K-P4-W	20 min
W2K-P4-X	20 min
W2K-P4-Y	20 min
W2K-P4-Z	20 min
W2K-P1-B	20 min
W2K-P1-C	20 min
W2K-P1-D	20 min
W2K-P1-E	20 min
W2K-P1-G	20 min
W2K-P1-H	20 min
W2K-P1-J	20 min
W2K-P1-K	20 min
W2K-P1-L	20 min
W2K-P1-M	20 min
W2K-P1-N	20 min
W2K-P1-U	20 min
W2K-P1-W	20 min
Unused Pins *	
Ground (external skin)	
W2K-P3-A	20 min
W2K-P3-B	20 min
W2K-P3-C	20 min
W2K-P3-D	20 min
W2K-P3-E	20 min
W2K-P3-F	20 min
W2K-P3-G	20 min
W2K-P3-H	20 min
W3K-P2-A	20 min
W3K-P2-B	20 min
W3K-P2-C	20 min
W3K-P2-E	20 min
W3K-P2-F	20 min
W3K-P2-G	20 min
W3K-P1-M	20 min
W3K-P1-K	20 min
W2K-P3-J	20 min
W3K-P1-F	20 min
W2K-P3-N	20 min
W2K-P3-L	20 min

* W2K P1-R,S,Z,c; W2K P2-F,J,N,R,U,a,b,s,t; W2K P5-E,H, J,K; W3K P1-1,N; W3K P2-D,H,J,K; (W2K P4-H,J,K,L,M,N, Q,R,S,U,V;) W2K P1-a,b,
*All these pins may be connected together and tested as one pin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.8 Cable capacitance. The maximum allowable capacity between G and a of W2K-P1 and pins B and b of W2K-P1, measured at 1000 Hz input voltage shall be 300 micro-microfarads when tested as specified in 4.4.5.

3.9 Environmental testing. The assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assembly shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.6. Tests shall be conducted in the order specified.

3.9.1 Waterproofness. The assembly shall not allow visible leakage inside the assembly when subjected to water spray simulating heavy rain for a period of 15 minutes when tested as specified in 4.4.6.1.

3.9.2 Temperature shock. The assembly shall withstand temperature shock between minus (-) 65 degrees Fahrenheit (°F) plus or minus (±) 5°F and plus (+) 160°F ± 5°F without evidence of damage to protective finishes, or any permenent warpage, and shall comply with 3.5.1, 3.6, 3.7 and 3.8 after being tested as specified in 4.4.6.2.

3.9.3 Humidity. The assembly shall withstand exposure to a relative humidity of 95 ± 5 percent at temperatures cycling from 80 ± 20°F to 160 ± 5°F to 80 ± 20°F without evidence of deterioration. The protective finish shall show no evidence of flaking or adhesion impairment. The assembly shall comply with the requirements of 3.6, 3.7 and 3.8 after being tested as specified in 4.4.6.3.

3.9.4 Low temperature cycling. The assembly shall withstand without damage, the low temperature and frost conditions specified in 4.4.6.4. The assembly shall comply with the requirements of 3.10, 3.6, 3.7 and 3.8 after being tested as specified in 4.4.6.5.

3.9.5 Vibration. The case section, ballistic, center, aft when mounted as part of the 8826411 assembly or mounted to the equivalent in weight, balance and mounting surfaces shall withstand flight vibration without damage along the geometric (flight) axes over a frequency range of 75-1000-75 Hz at a peak gravity (g) level of ± 5 g's (± 0.10 percent tolerance). The vibration shall be applied to the assembly from low to high and back to low frequency over a period of 0.9 ± 0.1 minute. There shall be no evidence of continuity of more than 20 microseconds duration between pins W3K-P1-C, H, and J connected together, and the case during vibration. The assembly shall comply with 3.10 after being tested as specified in 4.4.6.5.

3.9.6 Handling shock and vibration. The assembly when mounted as part of the 8826411 assembly or mounted to the equivalent in weight, balance and mounting surfaces shall withstand, without damage, the applicable requirements of the shock test of MIL-S-48512 when tested as specified in 4.4.6.6. Upon completion of this test the assembly shall comply with the requirements of 3.5.1 and 3.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.10 Deterioration. Any internal or external components which show evidence of deterioration or corrosion shall provide reason to consider the assembly as having failed to meet the test to which it was subjected.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwg 8825072) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Table III

First article inspection and tests

Inspection or Test	Units 1,2	Units 3,4,5	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X		X
2. Resistance (see 3.6)	X	X		X
3. Insulation resistance (see 3.7)	X	X		X
4. Mating with adaption kit (see 3.5.1)	X	X		X
5. Warhead compatibility (see 3.5.2)	X	X		X
6. Waterproofness (see 3.9.1)	X	X		X
7. Cable capacitance (see 3.8)	X	X		X
8. Handling shock and vibration (see 3.9.6)	X	X		X
9. Temperature shock (see 3.9.2)	X	X	X	
10. Humidity (see 3.9.3)	X	X	X	
11. Low temperature cycle (see 3.9.4)	X			X
12. Vibration (see 3.9.5)	X	X	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer of raw material or components, one production method or manufacturing operation. The entire lot shall be submitted as a unit for inspection.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Skin middle section (see dwg 8825095).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper diameter at station(s) 100.0, 80.0, 64.0, 48.7, 40.7, and 32.7	Gage
	102. Improper skin contour	Gage
	103. Improper length	Gage
	104. Improper welding (dye penetrant required)	Visual/Manual
	105. Improper paint coverage (applicable when assembly is procured as a separate item)	Visual
	106. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.3.2.1.2 Case section, ballistic, center, aft (see dwg 8825072).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Concentricity at station(s) 100.0, 80.0, 64.0, 48.7, and 32.7	TIR
2.	Perpendicularity of forward face at station 32.5 with functional centerline	TIR
3.	Mismatch at longitudinal joints between doors and skin	Gage
4.	Mismatch at transverse joint between doors and skin, step up and step down	Gage
Major:		
101.	Location of 4 slots and notches at station 32.50	Gage
102.	Location of 4 slots at station 32.50	Gage
103.	32 holes at station 100.0 location, size and size of countersinks incorrect	Gage
104.	Rivet heads exceeding ± 0.015 from skin surface	Gage
105.	Improper location and clamping of cables	Visual
106.	Improper connection of cables to fuze rings	Visual
107.	Fuze rings #2, #3, and #4 improper located and fastened	Visual
108.	Door gasket cemented out of position	Visual
109.	Improper metal finish (interior and exterior)	Visual
110.	Evidence of poor workmanship (see 3.11)	Visual
111.	Parts missing, damaged or loose (special attention for lockwashers missing on cable brackets and cable guards)	Visual/Manual
112.	Door hinges and latch function improperly	Manual
113.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Minor:

201. Dents (0.050 inch maximum depth, concave, with a maximum diameter of 2 inches, no limitation of spacing).....Gage

202. Scratches (0.010 to 0.030 inch depth shall not exceed 10 inches in length, scratches of less than 0.010 inch in depth, no limitation in length; no limitation of width).....Gage

4.3.2.1.3 Box, packing, ammunition, for case section assembly, ballistic, center, aft prior to closing (see dwg 8822030).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Pad missing or incorrect.....Visual

102. Connector not wrapped or wrapped improperly.....Visual

103. Humidity card missing.....Visual

104. Case section incorrectly positioned....Visual

105. Desiccant bag missing or mislocated....Visual

106. Excess air within bag.....Visual

107. Cable ends not tied securely.....Visual/Manual

108. Cleats missing, loose or mislocated....Visual/Manual

109. Supports loose, missing or incorrect...Visual/Manual

110. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.2.1.4 Box, packing, ammunition, for case, section assembly, ballistic, center, aft (see dwg 8822030).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Board broken, cracked or split.....Visual

102. Skid missing or incorrect.....Visual

103. Marking missing or incorrect.....Visual

104. Evidence of poor workmanship (see 3.11).....Visual

105. Strapping missing, mislocated, loose or incorrect.....Visual/Manual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Major: (Con't)
106. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Resistance	Major
b. Insulation resistance	Major
c. Cable capacitance	Major
d. Handling shock and vibration	Major
e. Vibration	Major

Of these tests, (e) is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Resistance, insulation resistance and cable capacitance. Each assembly in the lot shall be tested in accordance with 4.4.3 and 4.4.4. If any item fails to comply with 3.6 and 3.7, the item shall be rejected and removed from the lot.

4.3.2.2 Handling shock and vibration and vibration. A sample shall be selected at random in accordance with Table VI. The sample shall be tested in accordance with 4.4.6.6 and 4.4.6.5. If any item fails to comply with 3.9.6 and 3.9.5, the lot shall be rejected.

Table VI

Lot Size	Sample Size
2-180	2
181-500	3
501-800	5
801-1300	7

Note: For destructive tests:
Lot Size Required = Number Required to be Delivered + Sample Size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions designed or specified by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I as specified on DI-R-1714 (see 6.1.2).

4.3.4.2 Minor characteristics and tests. The acceptance inspection equipment, operating isntructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating with adaption kit assembly. The case section, ballistic, center, aft assembly shall be mounted to mating assemblies or a fixture equivalent to the mating surfaces on Drawing 8825153.

4.4.2 Warhead compatibility. When testing for warhead assembly requirements, the functional cargo, described in Drawing 8831129 Mark 31 Warhead Compatibility Information or inert equivalent shall be used. The functional cargo shall be capable of being assembled with the case section, ballistic, center, aft without distortion or damage of any parts and without the use of excessive force.

4.4.3 Resistance. A resistance measuring instrument shall be connected to the terminals specified in Table I for measurement of circuit resistance.

4.4.4 Insulation resistance. Insulation resistance shall be measured by means of a suitable instrument at the terminals specified in Table II. The test instrument shall be current limited to 5 milliamperes to protect human life. Connector shells shall be grounded for this test. Unused pins may all be connected together and tested as one terminal.

4.4.5 Cable capacitance. The capacitance between pins G and a of W2K-P1 and pins B and b of W2K-P1 shall be measured with a suitable capacitance measuring instrument.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.4.6 Environmental testing (see 3.9).

4.4.6.1 Waterproofness. The assembly shall be mounted horizontally and impacted with a water spray directed within 15 degrees of vertical. The rain test chamber temperature shall be maintained at 80 + 20°F throughout the test period. A simulated rain fall of 10 + 2 inches per hour shall be produced by means of a water spray nozzle. The rainfall shall be dispersed uniformly over the test area. During the test period the assembly shall be rotated once about the longitudinal axis. Following the exposure period the assembly shall be removed from the test chamber, wiped dry externally, disassembled from the rest of the 8825153 assembly and checked for water leakage.

4.4.6.2 Temperature shock. The assembly shall be placed within a test chamber where the temperature is -65 + 5°F. The assembly shall be subjected to this temperature for a period of 4 hours minimum at the conclusion of which, and within 5 minutes, the assembly shall be transferred to a chamber having an internal temperature of 160 + 5°F. The item shall be subjected to this temperature for a period of 4 hours minimum. This completes one cycle. The number of complete cycles shall be three. At the conclusion of the third cycle the item shall be removed from the test chamber and within a period of 30 minutes shall be tested as specified in 4.4.1, 4.4.3, 4.4.4 and 4.4.5.

4.4.6.3 Humidity. The assembly shall be placed in the test chamber mounted to simulate the installed conditions. The chamber temperature shall be 80 + 20°F. During the first 2 hour period, the temperature shall be gradually raised to 160 + 5°F. The temperature of 160 + 5°F shall be maintained during the next 6 hour period. During the following 16 hour period, the temperature in the chamber shall be gradually reduced to 80 + 20°F which completes one cycle. The relative humidity throughout the cycle shall be 95 + 5 percent. The test shall consist of 15 cycles (360 hours). Upon completion of the 15th cycle, the assembly shall be removed immediately from the test chamber and excessive moisture removed by blowing dry filtered air over the unit for a period of 5 minutes. The assembly shall then be tested as specified in 4.4.3, 4.4.4 and 4.4.5.

4.4.6.4 Low temperature cycle. The assembly with electrical connections protected against frost shall be placed in the test chamber and set up to simulate installed conditions. The chamber temperature shall be 80 + 20°F. During the first 2 hour period the temperature shall be gradually reduced to -65 + 5°F. The temperature of -65 + 5°F shall be maintained for 4 hours minimum. During the following 2 hour period the temperature shall be gradually increased to 80 + 20°F. The test chamber shall be maintained in this condition for 4 hours minimum. This completes one cycle. The assembly shall be subjected to 10 cycles. The assembly shall then be tested as specified in 4.4.3, 4.4.4 and 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.4.6.5 Vibration. The assembly shall be mounted to a vibration producing apparatus placed in an ambient temperature of 80 ± 20°F and vibrated along the geometric (flight) axis of the assemblies at the frequencies and accelerations and for the time durations specified. The accelerations specified shall be determined by suitable measuring devices attached to the mounting surface at Station 115 at the point of attachment. The monitoring circuit shall be capable of measuring circuit closure of 20 microseconds duration. The assembly shall then be inspected as specified in 4.3.2.1.

4.4.6.6 Handling shock and vibration. The assembly shall be mounted in a shipping container case, Atomic Warhead Section 8839515 or equivalent. The mounted assembly shall be tested in accordance with the shock test of MIL-S-48512. Following these tests, the assembly shall be inspected as specified in 4.4.1 and 4.4.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be packaged in accordance with dwg 8822030 using the specified moisture barrier.

5.1.2 Level C. The assembly shall be packaged in accordance with dwg 8822030 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The assembly shall be packed in accordance with dwg 8822030.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822030.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1679 dated 6 September 1961 With Amendment 1 Dated 8 December 1961

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No.: 1115-A049

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.